GreenEquip: Decentralized Farm Machinery Rental Application using Blockchain

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled **"GreenEquip: Decentralized Farm Machinery Rental Application using Blockchain"** is a bonafide work of **Sidra Khan (22106028), Om Panchal (22106025), Devang Dhuri (22106036), Sumant Bhise (22106095)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning).**

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Project Report Approval for B.E.

This project report entitled ***"GreenEquip: Decentralized Farm Machinery Rental Application using blockchain"*** by ***"Author Name"*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning), 2025-26***.

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Abstract

GreenEquip is a decentralized application (dApp) that operates on the public Ethereum blockchain, creating a trustless and efficient peer-to-peer marketplace for the rental of agricultural equipment. By leveraging blockchain technology, the platform removes the need for traditional intermediaries, connecting equipment owners directly with renters in a transparent and secure environment.

The backend logic is encapsulated in the `FarmChain.sol` smart contract, written in the Solidity programming language. This contract defines the core data structures, such as `Equipment` and `Rental`, and the business logic for all interactions. Key functions include `registerEquipment`, which allows owners to immutably record their assets on the blockchain, and `rentEquipment`, which automates the rental process, including the secure transfer of funds from the renter to the owner using Ether as the native currency. All state changes, such as new registrations or rentals, trigger events that provide a verifiable and public audit trail.

The user interface is a modern, responsive web application built with HTML, Tailwind CSS, and JavaScript, which serves as the primary entry point to the Web3 ecosystem. It uses the Ethers.js library as a bridge to communicate with the Ethereum network. This allows the frontend to read data from the `FarmChain.sol` contract (e.g., fetch equipment listings) and to write data to it by prompting users to sign transactions with their MetaMask wallet. This architecture ensures that users are always in control of their own funds and data, interacting directly with a self-executing and tamper-proof digital agreement.

Keywords: Decentralized Application (dApp), Smart Contracts, Ethereum, Blockchain, Peer-to-Peer Marketplace.

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Chapter 1

Introduction

The agricultural sector remains the cornerstone of the global economy, providing sustenance and livelihood for a significant portion of the world's population. However, despite its fundamental importance, the industry, particularly in developing nations, is fraught with systemic inefficiencies, inequities, and a lack of transparency that hinder its growth and sustainability. Smallholder farmers, who form the backbone of this sector, often face considerable barriers in accessing essential resources, including modern agricultural equipment and affordable credit. Traditional systems for equipment rental are frequently centralized, geographically fragmented, and plagued by intermediaries who add layers of cost and complexity, ultimately reducing the farmers' profitability and productivity. Farmers often have to undertake arduous journeys to rental hubs with no guarantee of equipment availability or fair pricing, a process that is both time-consuming and financially draining. This lack of an efficient, trustworthy marketplace not only impedes the adoption of modern farming techniques but also perpetuates a cycle of low yields and economic vulnerability. The urgent need for a transformative solution that can democratize access to resources, foster trust, and streamline operations has led to the exploration of emerging technologies to modernize this vital industry.

In recent years, blockchain technology has emerged as a powerful paradigm with the potential to revolutionize various sectors by establishing decentralized, transparent, and immutable systems for value exchange and data management. Originally developed as the foundational technology for cryptocurrencies, its applications now extend far beyond finance, offering robust solutions for supply chain management, asset tracking, and governance. A blockchain is, at its core, a distributed ledger that records transactions in a secure and tamper-proof manner across a network of computers, eliminating the need for a central authority to validate or mediate interactions. This decentralized architecture, combined with the power of smart contracts—self-executing agreements with predefined rules encoded directly onto the blockchain—provides a framework for creating trustless peer-to-peer networks. Within the agricultural domain, this technology offers a compelling solution to the aforementioned challenges. By creating a transparent and auditable record of all transactions, blockchain can enhance food safety, improve supply chain traceability, and, crucially, facilitate the creation of decentralized marketplaces for agricultural assets and services. It is within this context that the GreenEquip project was conceived. This

project presents a full-stack decentralized application (dApp) that leverages the Ethereum blockchain to create a peer-to-peer marketplace specifically for the rental of agricultural equipment, aiming to directly connect equipment owners with farmers in a secure, transparent, and efficient manner.

The GreenEquip dApp is architected as a complete Web3 solution, integrating a sophisticated backend powered by a Solidity-based smart contract with a modern, user-friendly frontend. The smart contract, GreenEquip.sol, serves as the immutable business logic of the application, programmatically defining the rules for equipment registration, rental agreements, payment processing, and equipment returns. This removes the need for human intermediaries, as all transactions are automatically executed and verified by the network once the predefined conditions are met. The frontend, built with HTML, Tailwind CSS, and JavaScript, provides an intuitive interface for users to interact with the dApp. It utilizes the Ethers.js library to communicate with the Ethereum blockchain, allowing users to connect their MetaMask wallets, browse equipment listings, initiate rentals, and view transaction histories in a seamless manner. The entire development process is managed within the Hardhat environment, a professional framework for Ethereum development that provides robust tools for compiling, testing, and deploying smart contracts, ensuring the reliability and security of the application.

The significance of the GreenEquip project extends beyond its immediate practical application. While much of the existing research on blockchain in agriculture focuses on supply chain traceability and food safety, this project addresses a distinct gap by creating a functional, decentralized marketplace for asset sharing and rental—a critical need for smallholder farmers. It serves as a practical implementation of the theoretical models proposed in academic literature, demonstrating a viable solution to the challenges of equipment accessibility. Furthermore, this repository showcases a broader range of technological capabilities through two supplementary, standalone applications. The first, FileDrop, is a client-side file management tool that highlights modern frontend development techniques, including drag-and-drop functionality and local storage persistence. The second, an AI Essay Grader, integrates the Gemini API for advanced natural language processing and Chart.js for data visualization, demonstrating the potential of AI in educational applications. In its entirety, this project not only delivers a robust decentralized solution for a real-world problem but also serves as a comprehensive educational resource, illustrating the powerful convergence of blockchain, modern web development, and artificial intelligence. This report will provide a detailed examination of the system's architecture, technical implementation, and functionalities, followed by an analysis of the results and a discussion of future work

Chapter 2

Literature Survey

1. **Smart Farming: Securing Farmers Using Block Chain Technology and IOT (Praveen Pappula, Mohammed Ali Shaik, Sampath Kumar, Tanupriya Choudhury, June 2021):** I am reviewing this paper to understand how blockchain enables a transparent and secure machinery rental system, which is the core of my project. It explains how a web platform and blockchain backend can manage usage logs, directly supporting the architecture of my GreenEquip dApp. The insights validate my use of smart contracts for handling rental agreements and transactions.
2. **Block chain based agricultural supply chain-A review (Harshitha M S, Shashidhar R 2021):** This paper reviews blockchain's role in the secure tracking and management of financial and digital assets. I connect its concepts to my project by treating farm equipment as manageable assets whose ownership and rental status must be transparently tracked. This strengthens the asset registration and management functions within my GreenEquip.sol contract.
3. **Rental And Loan System In Agriculture Using Blockchain Technology (Bhavani Rathod, Zenodo (CERN European Organization for Nuclear Research), 2023):** I am reviewing this for its framework on securing agricultural data using blockchain, even without its deep learning components. It demonstrates how blockchain ensures trust and integrity for farm-related records, like rental history and equipment details. I use this to frame the importance of the secure, on-chain ledger in my GreenEquip system.
4. **An E-Commerce Platform Of Agriculture Products Using Blockchain Technology (RMDSSOE Savitribai Phule Pune University, Pune, India, 2023):** This review provides a broad perspective on using smart contracts and blockchain to solve agricultural supply chain challenges like poor data integration. It highlights the importance of traceability and farmer-centric applications. I use it as a foundational source to justify the motivation and background for my project's goal of creating a more transparent rental market.
5. **Farming Equipment Rental System (Omkar Jadhav, Chaitanya Gamane, Shraddha Jagtap, Paras Mahale, 2025):** This study presents blockchain for securing agricultural transactions and improving supply chain logistics. It emphasizes how decentralization removes dependence on central servers, which

is a key principle of my dApp. I relate its findings to building trust between equipment owners and renters on the GreenEquip platform.

6. **From Fields to Sharing Yields: A Novel Approach to Affordable Farm Equipment Access (Proceedings of the International Conference on Innovative Computing & Communication (ICICC), 2024):** This paper focuses on optimizing systems for agricultural equipment, which aligns with my project's domain. It shows the importance of having an efficient backend to handle complex scheduling and data. I link this to the need for a robust and scalable smart contract design in GreenEquip to manage multiple rental requests effectively.
7. **Blockchain and IoT in Smart Agriculture: Analysis, Opportunities, Challenges, and Future Research Directions (Fatma MARZOUGUI, Mohamed ELLEUCH, and Monji KHERALLAH. 2024):** I am reviewing this to explore how smart contracts can automate financial services for farmers, such as insurance or loans. It demonstrates a practical use case for blockchain in agricultural finance. I connect this concept to the potential future expansion of GreenEquip to offer integrated loan or rental insurance features for its users.

Year, Publication	Title	Domain	Algorithm	Dataset Used	Research Gap
2025	Smart Farming: Securing Farmers Using Block Chain Technology and	Agriculture (Equipment Sharing)	Blockchain backend + Web platform	Farm machinery usage logs LINK	Lack of real-world multi-stakeholder adoption and IoT-enabled predictive maintenance validation.
2024	Block chain based agricultural supply chain-A review	Finance / Asset Management	Blockchain + Smart Contracts	Financial assets (stocks, bonds, real estate, digital assets) LINK	Interoperability, energy efficiency, and regulatory compliance

					remain unresolved.
2023	Rental And Loan System In Agriculture Using Blockchain Technology X	Smart Farming	Deep Learning (SAE) + Blockchain (ePoW)	IoT farm sensor data (simulated) LINK	Real-time deployment suffers from latency, limited edge optimization, and simulated datasets.
2021	An E-Commerce Platform Of Agriculture Products Using Blockchain Technology X	Agriculture & Blockchain	Blockchain (Smart Contracts, Traceability)	Agricultural supply chain case studies LINK	Poor ERP integration, fragmented data sources, and low farmer usability hinder adoption.
2021	Farming Equipment Rental System	Agri-Food Supply Chain	Blockchain + IoT	Food supply chain logistics data	Dependence on centralized servers and absent IoT interoperability standards undermine decentralization .
2020	From Fields to Sharing Yields: A Novel Approach to Affordable Farm Equipment	Agricultural Equipment Manufacturing Optimization	Dynamic Artificial Ant-Bee colony(hybrid Metaheuristic	Simulated manufacturing service scheduling data LINK	Simulation-heavy approach without field validation, cybersecurity

	Access (Proceedings of the International Conference on Innovative Computing & Communication				modeling, or heterogeneous IoT data handling.
2019	Blockchain and IoT in Smart Agriculture: Analysis, Opportunities, Challenges, and Future Research Directions	Agri-Insurance	Smart Contracts on NEO Blockchain	Weather & socio-economic data (SE Asia) LINK	Geographic limitation, data reduce reliability.

Chapter 3

Limitations of Existing Systems

The existing farm machinery rental ecosystem suffers from a number of structural and operational inefficiencies that reduce its effectiveness, especially for small and marginal farmers. A major limitation is the high transaction cost imposed by intermediaries and brokers, who charge heavy commissions and inflate rental prices, reducing the economic feasibility of equipment sharing. In addition, there is a serious lack of trust and transparency, since most agreements are made informally without proper contracts or verification, leading to frequent disputes over payment delays, usage time, damages, or accountability. The problem is worsened by the absence of standardized pricing models, with charges often determined arbitrarily by machinery owners or intermediaries, making modern machinery unaffordable for many farmers. Accessibility is another critical issue, as rural and remote farmers struggle to gain access to advanced machinery due to weak distribution networks, poor infrastructure, and lack of awareness. Moreover, once machinery is rented, there is no real-time monitoring of usage, which creates opportunities for misuse, unauthorized sub-leasing, and delayed returns.

These issues are compounded by the reliance on manual, paper-based records, which are highly vulnerable to tampering, fraud, and mismanagement, further reducing accountability and reliability. The dominance of middlemen exacerbates exploitation, as brokers not only inflate costs but also create dependency by controlling access to machinery. Collectively, these challenges make existing systems fragmented, exploitative, and inefficient. They demonstrate the urgent need for a decentralized, transparent, and technology-driven rental ecosystem that minimizes costs, ensures trust through verifiable contracts, & enables real-time monitoring.

Chapter 4

Problem Statement, Objectives and Scope

Problem Statement

Modern farming faces challenges of high labor costs, machinery scarcity, and inefficient resource allocation, especially for small and marginal farmers. The objective is to develop a decentralized farm machinery system that allows farmers to share and access farm equipment on demand, optimizing usage, reducing costs, and improving productivity. The system provides easy access to farm machinery, secure payments, and transparent records of all transactions. By leveraging blockchain and smart contracts, it ensures trust, automation, and tamper-proof agreements between farmers and lenders. This peer-to-peer approach reduces intermediaries, encourages fair practices, and creates a sustainable ecosystem for agricultural machinery sharing. Over time, it can improve financial inclusion, empower smallholders, and enhance overall farming efficiency.

Objectives

1. **To Develop a Decentralized, Blockchain-Based Platform:** To design and implement a fully functional decentralized application (dApp) on an Ethereum-based network that allows users to directly interact with a shared ledger for renting agricultural equipment, removing the need for traditional intermediaries.
2. **To Automate Rental Transactions Using Smart Contracts:** To create and deploy a secure Solidity smart contract that automates the entire equipment rental lifecycle, including equipment registration, peer-to-peer rental agreements, and the secure transfer of payments from renter to owner.
3. **To Build an Intuitive and Feature-Rich User Interface:** To construct a comprehensive user-facing web interface that enables seamless interaction with the blockchain, allowing users to connect their crypto wallets, register new equipment with detailed information, and easily view, filter, and rent available machinery.
4. **To Establish a Transparent and Efficient Agricultural Marketplace:** To create a trusted and transparent marketplace that solves key challenges for farmers by providing verifiable ownership, transparent rental terms, and direct, secure transactions, thereby reducing costs and increasing access to essential agricultural equipment.

Scope

GreenEuip dApp has created a decentralized marketplace for agricultural equipment rentals with plans to grow into a vital agricultural tool. Next steps include adding a blockchain-based reputation system for transparent user reviews, and creating verified digital identities for equipment with on-chain ownership, rental, and maintenance histories to support a trusted secondary market. The platform aims to integrate smart contract insurance triggered by external data oracles, automating payouts for events like droughts. Long-term, GreenEuip envisions evolving into a full decentralized agricultural finance and supply chain platform, enabling credit scoring based on rental history, facilitating loans, and providing end-to-end product traceability to ensure authenticity and access premium markets. This will transform GreenEuip into a cornerstone of the digital agricultural economy.

Chapter 5
Proposed System

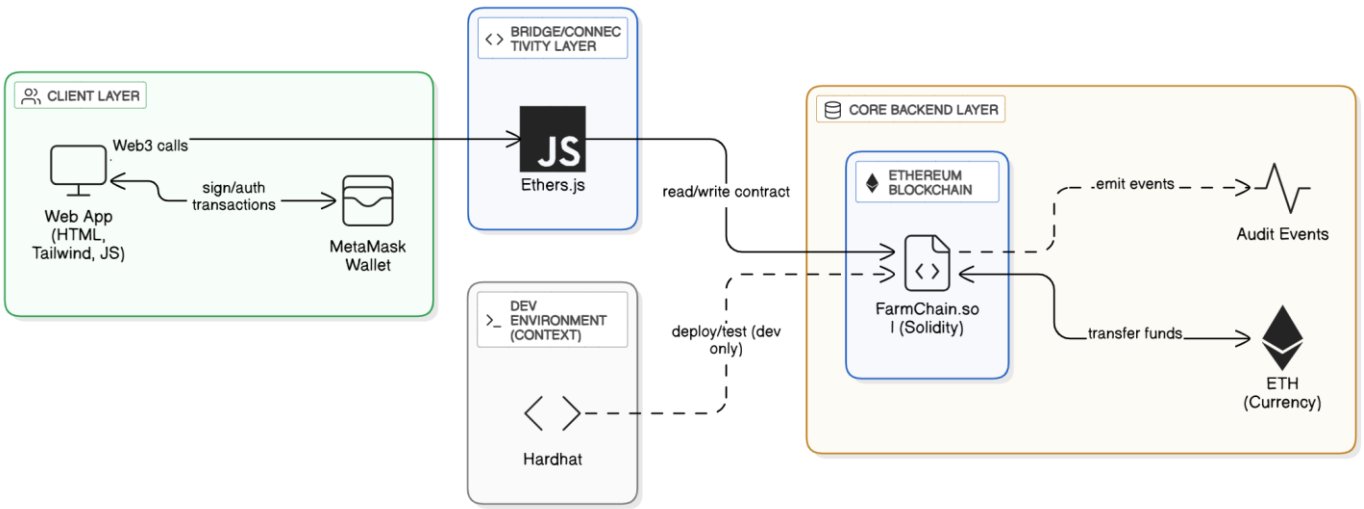


Figure 1.1 Architecture Diagram for GreenEquip

This diagram illustrates the workflow of a GreenEquip This diagram represents a decentralized equipment rental system built using Web3 (blockchain) technology. It is divided into three main layers: User Interface (Web2), Connectivity & Bridge, and Blockchain & Backend Logic (Web3).
The User Interface layer, which is the Web2 client-side, allows the farmer (whether lender or renter) to interact with the system. This layer includes mobile and web interfaces built with HTML, Tailwind CSS, and JavaScript. Through this interface, users can list their equipment for renting, browse or search available equipment, book the selected equipment, track the status of transactions such as rents or disputes, and connect their cryptocurrency wallets like MetaMask. All actions such as listing, booking, or searching prepare a transaction to be sent to the blockchain, while tracking functionality fetches events to update the interface in real time.

The Connectivity & Bridge layer acts as the link between the Web2 frontend and the Web3 blockchain. Ethers.js functions as the bridge, preparing and sending transactions to the blockchain, fetching blockchain events to update the user interface, and prompting MetaMask for transaction signatures. MetaMask handles signing transactions, managing user authorization, and overseeing wallet management. When a user performs an action, the interface prepares a transaction which is sent to Ethers.js, MetaMask signs it, and then the signed transaction is sent to the blockchain. The blockchain events are subsequently fetched by Ethers.js to update the UI.
The Blockchain & Backend Logic layer handles decentralized storage, smart contracts, and payment flows. Transactions are stored on the Ethereum public blockchain in a distributed ledger. FarmChain.sol, a smart contract written in Solidity, manages the registration of equipment, renting, payments, and dispute resolution. Hardhat is used as a development tool for deploying or testing these

smart contracts. The payment flow ensures that Ether transfers, security deposits, and disputes are managed automatically. Transactions from MetaMask reach FarmChain.sol, which writes and reads state while emitting events to the blockchain ledger. The smart contract manages all payments and dispute resolutions without centralized intervention.

Overall, the system works as follows: the farmer interacts with the Web2 interface, which prepares blockchain transactions. Ethers.js acts as the bridge to MetaMask for signing, after which the signed transactions are sent to the Ethereum blockchain. The smart contract executes the required logic, updates the ledger, and manages payments, while blockchain events are fetched back to update the interface in real time. The system ensures that the Web2 frontend allows smooth user interaction, the Web3 bridge facilitates secure blockchain communication, and smart contracts automate logic, security, and payments, making the platform decentralized, trustless, and transparent.

Algorithm:

Key Components of the Proposed Framework:

The GreenEquip dApp is built on a multi-layered architecture that separates on-chain and off-chain responsibilities to ensure security, scalability, and a rich user experience. The framework is composed of three primary layers: the blockchain and smart contract layer, the frontend/presentation layer, and the intelligent agents and chatbot layer.

Blockchain and Smart Contract Layer (On-Chain Core)

This layer forms the decentralized backend and acts as the single source of truth for the entire application. It is responsible for all state changes and value transfers within the ecosystem.

1. Ethereum blockchain: The public, permissionless blockchain that provides a secure and immutable ledger for all transactions.
2. GreenEquip.sol smart contract: The core of the dApp, written in Solidity, which programmatically enforces the business logic of the equipment rental marketplace. It manages equipment registration, rental agreements, secure payment transfers, and status updates for all assets.
3. Transaction and event logs: All interactions with the smart contract are permanently recorded as transactions on the blockchain. The contract also emits events such as EquipmentRegistered and EquipmentRented, creating a transparent and auditable history of activities. These logs are essential for the off-chain layers to monitor and respond to changes in the system.

Frontend/Presentation Layer (Off-Chain Client)

This layer is the user-facing component of the dApp, responsible for providing an intuitive interface for users to interact with the blockchain backend. It runs entirely within the user's web browser.

1. Web interface (index.html): A single-page application built using HTML, Tailwind CSS, and JavaScript that provides the visual interface for browsing and renting equipment.
2. Web3 connectivity (Ethers.js): The Ethers.js library is used to connect the frontend with the Ethereum network. It integrates with the user's MetaMask wallet, retrieves live contract data, and prompts transaction signing for operations like renting or registering equipment.
3. Internationalization (i18next): The i18next framework enables multilingual support. With English and Hindi integrated, the application becomes more inclusive and accessible to a wider audience, particularly in agricultural regions where Hindi is widely spoken.

Intelligent Agents and Chatbot Layer (Off-Chain Client/Service)

This layer enhances the dApp by introducing intelligence, optimization, and interactive user support. It processes on-chain data and supplements it with off-chain logic to guide users.

1. Rental optimization agent: An off-chain agent that analyzes historical rental data to suggest dynamic pricing strategies to equipment owners and identify cost-effective rental options for farmers. This

service continuously reads from the blockchain and operates as a backend module.
2. Reputation management agent: A hybrid system that evaluates user trustworthiness by calculating reputation scores based on on-chain activity such as successful rentals. While the reputation scores are stored in a dedicated smart contract (Reputation.sol), calculations are performed off-chain to minimize gas fees.
3. Agricultural chatbot (agricultural-chatbot.js): A rule-based chatbot running client-side, designed without dependency on external APIs. It leverages a predefined dataset (farming-advice-dataset.js) to provide instant farming advice, support navigation, and answer equipment-related queries. Being state-aware, the chatbot can access live data from the blockchain, enabling it to respond accurately about equipment availability, rental pricing, and usage tips.

Analysis:

The proposed framework integrates blockchain technology with modern web development and intelligent agents to create a robust and user-centric decentralized marketplace. The analysis of this framework reveals several key strengths.
First, the architecture effectively leverages decentralization to build trust and transparency, which is a core challenge in traditional agricultural markets. By using the GreenEquip.sol smart contract as the sole arbiter of transactions, the system eliminates the need for intermediaries, reduces the risk of fraud, and ensures that all participants are operating on a level playing field with access to the same immutable data. This directly addresses the issues of untrustworthy parties and the lack of transparency in conventional supply chains, as highlighted in the supporting literature.
Second, the framework places a strong emphasis on user experience and accessibility, which is often a secondary consideration in many blockchain projects. The frontend is not merely a functional interface but is designed to be intuitive, responsive, and multilingual. The integration of the agricultural chatbot is a critical component of this strategy. By providing instant, context-aware assistance without requiring users to have a deep understanding of blockchain, the chatbot lowers the barrier to entry and makes the dApp more approachable for the target audience of farmers and equipment owners.
Third, the inclusion of intelligent agents demonstrates a forward-thinking approach that goes beyond simple transaction processing. The rental optimization and reputation management agents transform the dApp from a passive ledger into a dynamic marketplace. The optimization agent introduces market efficiencies by using historical data to inform pricing, while the reputation agent builds a layer of trust that is essential for a healthy peer-to-peer ecosystem. This aligns with academic research suggesting that blockchain's true potential is realized when combined with other technologies like IoT and data analytics to create smart agricultural systems.
Finally, the technical design is both practical and scalable. By keeping complex computations and large data storage off-chain (in the agents' backend and the chatbot's local dataset), the system avoids the high gas fees and performance limitations of the Ethereum blockchain. It uses the blockchain for what it does best: securing state and verifying transactions. The proposed architecture in "Agriculture-Food Supply Chain Management Based on Blockchain and IoT" supports this modular approach, advocating for multi-chain or layered systems to handle scalability and storage concerns, a principle that this framework embodies.

Chapter 6
Experimental Setup

Software Requirements: -

1. **OS**

   The development environment is flexible and can be configured on any of the major operating systems, including **Windows, macOS, or Linux**. Since the end-user application is **platform-agnostic**, it does not depend on the underlying operating system. Instead, it is accessed through a standard web browser, ensuring compatibility across devices such as desktops, laptops, and even tablets, as long as a supported browser is available.

2. **Web-Browser**

   A **modern web browser** with support for **JavaScript, Web3 integration, and browser extensions** is required for running the decentralized application (dApp). Recommended browsers include:
   - **Google Chrome**: Most widely used and has excellent compatibility with blockchain extensions.
   - **Mozilla Firefox**: Known for stability and strong developer support.
   - **Brave Browser**: Privacy-focused and comes with built-in Web3 support, making it particularly suitable for blockchain applications.

3. **MetaMask-Extension**

   The **MetaMask browser extension** is a mandatory requirement for all users who wish to interact with the Ethereum blockchain through the dApp. MetaMask acts as a secure bridge between the web application and the blockchain by providing:
   - **Wallet management** (creating and managing Ethereum accounts).
   - **Transaction signing** (confirming and executing blockchain operations).
   - **Network connectivity** (connecting the dApp to Ethereum mainnet, testnets, or custom networks like Hardhat).

4. **Node.js-Runtime-Environment**

   For the development setup, **Node.js** plays a crucial role. It provides the **JavaScript runtime** required for executing server-side scripts and managing blockchain project dependencies through **npm (Node Package Manager)**. Additionally, Node.js is necessary for:
   - Running the **Hardhat local blockchain network** for testing and debugging smart contracts.
   - Automating development workflows such as compiling, deploying, and verifying smart contracts.
   - Managing package dependencies and frameworks that form the backbone of the dApp development process.

Hardware Requirements: -

1. **Client Device (End-User)**
   - Any standard **desktop, laptop, or mobile device** capable of running a modern web browser.
   - No high-performance hardware is needed since the FarmChain dApp runs directly in the browser.

2. **Internet Connection**
   - A **stable broadband or mobile internet connection** is required.
   - Ensures smooth communication with the Ethereum blockchain for transaction processing and real-time data retrieval.

3. **Development Machine (Developer-Side)**
   - **Processor**: Dual-core CPU or higher.
   - **Memory (RAM)**: Minimum **4 GB** (8 GB recommended for optimal performance).
   - **Storage**: At least **10 GB free disk space** for dependencies, local blockchain data, and project files.
   - **Additional Tools**: Modern code editor (e.g., Visual Studio Code), Node.js runtime, and Hardhat local blockchain environment.

Operating Environment:

1. **Execution-Environment**

   The FarmChain application is designed to run primarily on the **client-side**, ensuring that users can interact with the platform without needing additional installations beyond a supported browser and the MetaMask extension. The following aspects define the execution environment:
   - The **frontend interface** (developed using modern web technologies) executes entirely within the user's **web browser**.
   - The **chatbot logic and interaction layer** also run client-side, leveraging browser-based JavaScript execution to deliver real-time responses and guidance to the user.
   - Since computation and logic are handled locally on the client's browser, system requirements for end-users remain lightweight.

   This client-side execution model not only improves performance and scalability but also reduces reliance on centralized servers, aligning with the decentralized nature of blockchain-based applications.

2. **Blockchain-Network**

   The core functionality of the FarmChain dApp relies on **Ethereum** or other **EVM-compatible blockchains**. This ensures flexibility and portability across multiple blockchain ecosystems.
   - **Deployment Environment**: In production, the application connects directly to the Ethereum mainnet or an EVM-compatible network to handle real-world transactions and smart contract execution.

- **Development & Testing Environment**: For developers, the system uses a **Hardhat Network**, a local Ethereum-like blockchain simulation. Hardhat provides:
  - Fast and reliable **local blockchain testing**.
  - Tools for **compiling, deploying, and debugging smart contracts**.
  - The ability to simulate real blockchain conditions (e.g., gas fees, block mining) without incurring actual costs.

Development Environment:

1. **Smart Contract Framework**

   - **Compiling** Solidity smart contracts into deployable bytecode.
   - **Deploying** contracts to local, testnet, or mainnet environments.
   - **Testing** contract logic with built-in testing tools (using JavaScript/TypeScript).
   - **Debugging and error tracing**, including stack traces, console.log outputs, and gas usage reporting.
   - **Simulating Ethereum nodes** locally through the Hardhat Network, enabling developers to test features without incurring real gas costs.

   This framework forms the backbone of the development workflow by ensuring reliable, repeatable, and cost-effective contract development.

2. **Smart Contract Language**

   Solidity is the programming language used for developing the smart contracts that govern the FarmChain dApp. Key contracts include:
   FarmChain.sol: The primary smart contract handling business logic and decentralized operations.
   Lock.sol: A supporting contract, often used in scaffolding or managing time-based operations such as escrow or staking.
   Solidity is specifically designed for the Ethereum Virtual Machine (EVM), making it the most suitable choice for ensuring secure and deterministic contract execution.

3. **Frontend Libraries**

   - **Ethers.js**

     A lightweight yet powerful JavaScript library designed for blockchain interactions. It is responsible for:
     - Connecting the frontend with **MetaMask** or other Ethereum wallets.
     - **Reading on-chain data** from deployed contracts.
     - **Sending transactions** and monitoring blockchain events.
     - Providing abstraction over low-level JSON-RPC calls to simplify development.

   - **Tailwind-CSS**

A **utility-first CSS framework** that allows developers to design responsive and futuristic UIs rapidly. Tailwind ensures:

- A modern, consistent look across devices.
- Faster UI development with pre-built utility classes.
- Easy customization and scalability of the design system.

- **i18next**

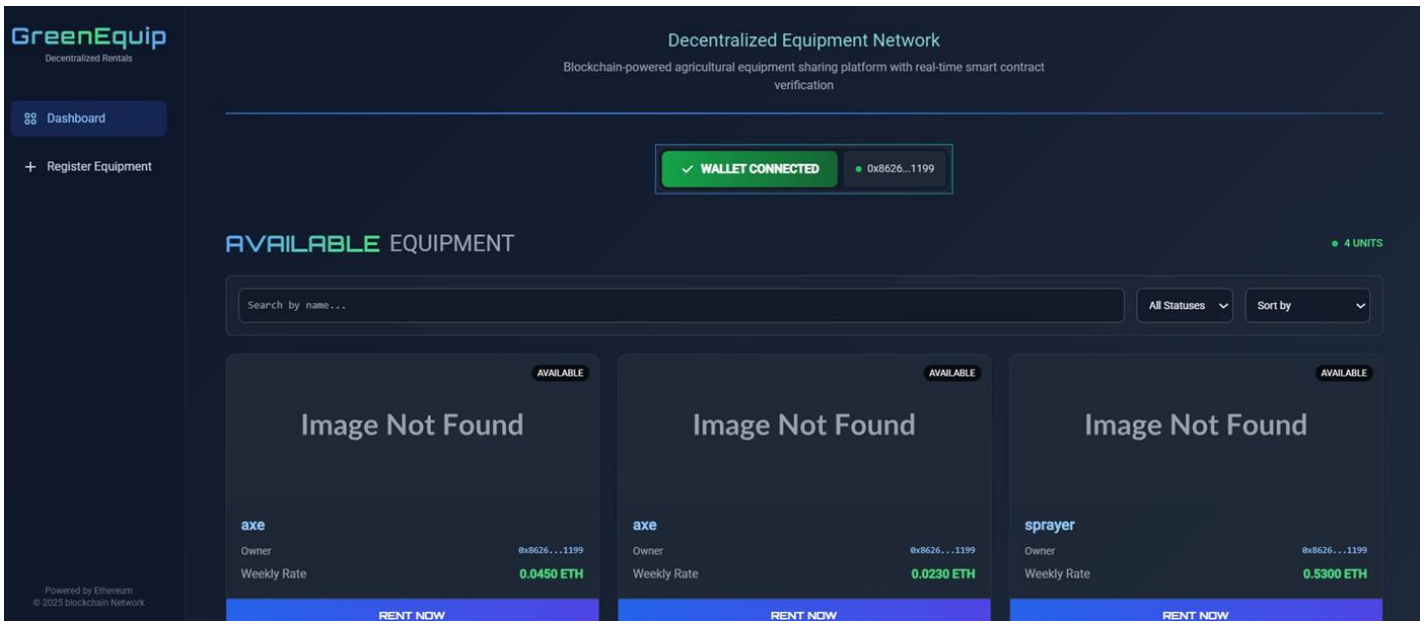
  An advanced **internationalization (i18n) library** for JavaScript applications. Within FarmChain, it is integrated to deliver:

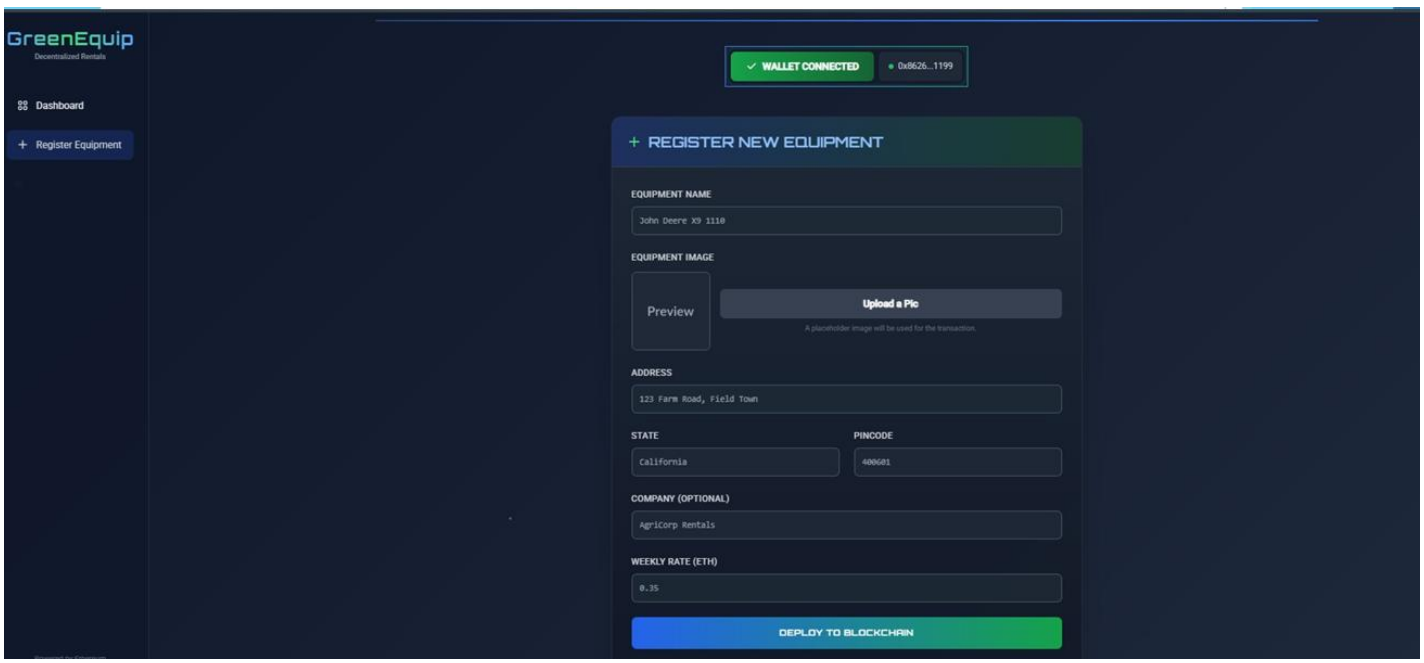
  - **Multilingual support**, currently for **English and Hindi.**
  - Seamless translation of the **user interface** and **chatbot responses**.
  - Enhanced accessibility for diverse user groups, making the platform more inclusive.

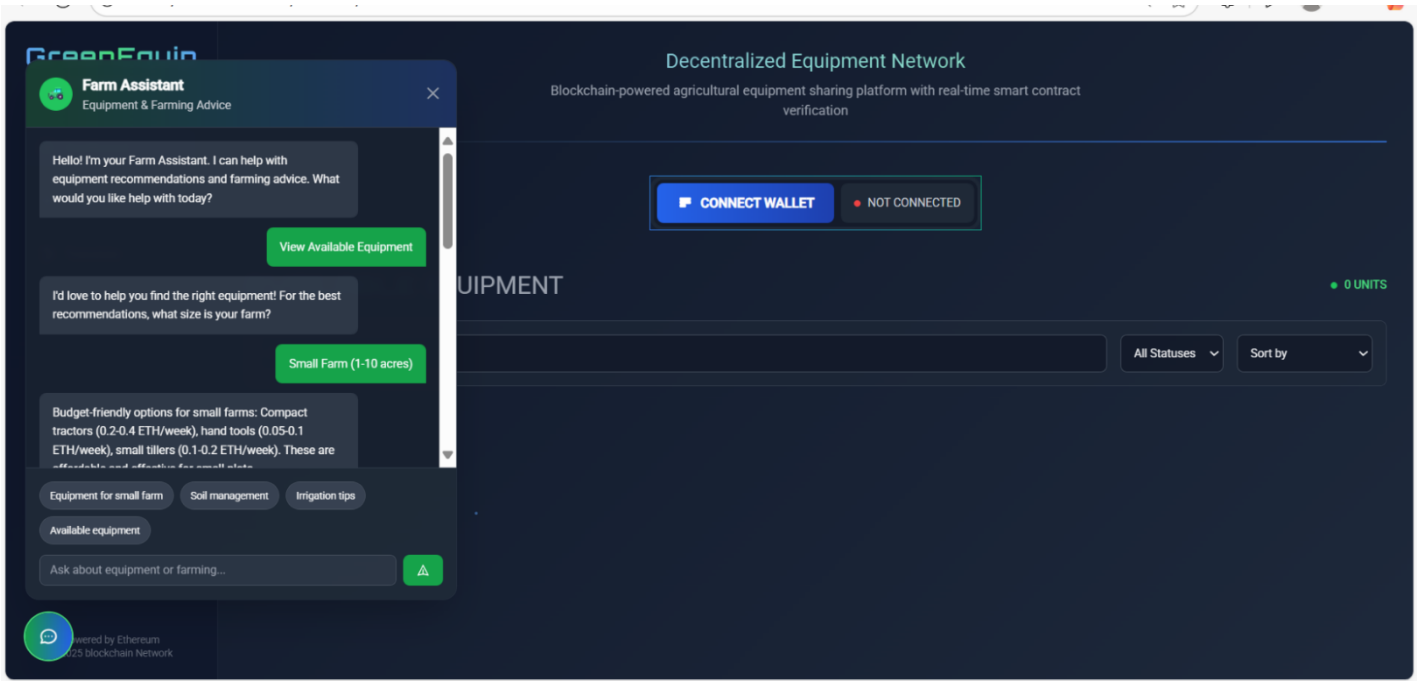
4. Code Editor

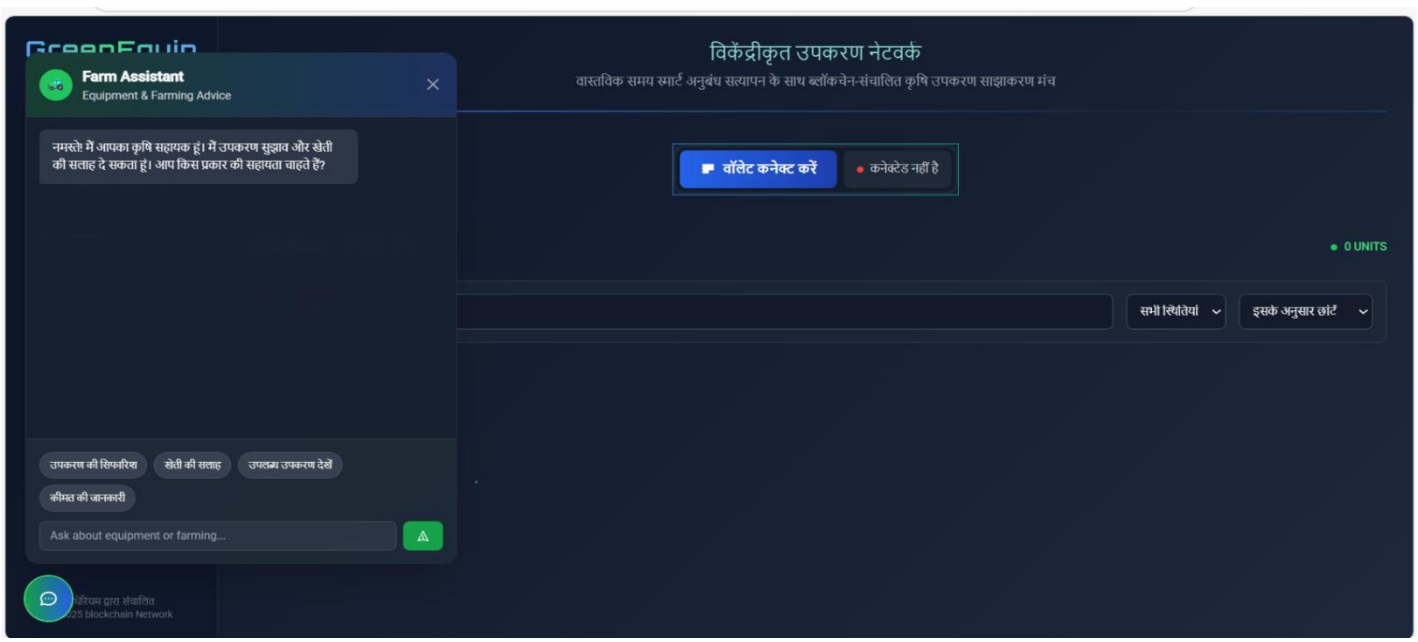
- Integrated support for JavaScript, TypeScript, and Solidity development.
- Extensions for Hardhat, Ethers.js, and Solidity debugging.
- Built-in Git integration for version control and collaborative workflows.

Expected Outcome






Farm Assistant
Equipment & Farming Advice
Hello! I'm your Farm Assistant. I can help with equipment recommendations and farming advice. What would you like help with today?
View Available Equipment
I'd love to help you find the right equipment! For the best recommendations, what size is your farm?
Small Farm (1-10 acres)
Budget-friendly options for small farms: Compact tractors (0.2-0.4 ETH/week), hand tools (0.05-0.1 ETH/week), small tillers (0.1-0.2 ETH/week). These are
Equipment for small farm
Soil management
Irrigation tips
Available equipment
Ask about equipment or farming...
Decentralized Equipment Network
Blockchain-powered agricultural equipment sharing platform with real-time smart contract verification
CONNECT WALLET
NOT CONNECTED
UIPMENT
0 UNITS
All Statuses
Sort by


Farm Assistant
Equipment & Farming Advice
नमस्ते! मैं आपका कृषि सहायक हूं। मैं उपकरण सुझाव और खेती की सलाह दे सकता हूं। आप किस प्रकार की सहायता चाहते हैं?
उपकरण की सिफारिश
खेती की सलाह
उपलब्ध उपकरण देखें
कीमत की जानकारी
Ask about equipment or farming...
विकेंद्रीकृत उपकरण नेटवर्क
वास्तविक समय स्मार्ट अनुबंध सत्यापन के साथ ब्लॉकचेन-संचालित कृषि उपकरण साझाकरण मंच
वॉलेट कनेक्ट करें
कनेक्टेड नहीं है
0 UNITS
सभी स्थितियां
इसके अनुसार छांटें

References

[1] Smart Farming: Securing Farmers Using Block Chain Technology and IOT (Praveen Pappula, Mohammed Ali Shaik, Sampath Kumar, Tanupriya Choudhury, June 2021): https://www.researchgate.net/publication/352377264_Smart_Farming_Securing_Farmers_Using_Block_Chain_Technology_and_IOT

[2] Dong, S., Abbas, K., Li, M., & Kamruzzaman, J. (2023). *SP2F:* Secure Privacy-Preserving Framework blending Deep Learning & Blockchain for Smart Farming. PeerJ Computer Science, 9, e1705: https://openknowledge.fao.org/server/api/core/bitstreams/66ce8120-ed1b-4469-a4ca-08529d3b0774/content

[3] Farming Equipment Rental System (Omkar Jadhav, Chaitanya Gamane, Shraddha Jagtap, Paras Mahale, 2025): https://www.irjmets.com/uploadedfiles/paper/issue_3_march_2025/68696/final/fin_irjmets1741518207.pdf

[4] Rental and Loan System In Agriculture Using Blockchain Technology (Bhavani Rathod, Zenodo (CERN European Organization for Nuclear Research), 2023): https://www.academia.edu/111652399/Rental_And_Loan_System_In_Agriculture_Using_Blockchain_Technology

[5] An E-Commerce Platform of Agriculture Products Using Blockchain Technology (RMDSSOE Savitribai Phule Pune University, Pune, India, 2023): https://ijcrt.org/papers/IJCRT2311460.pdf

[6] From Fields to Sharing Yields: A Novel Approach to Affordable Farm Equipment Access (Proceedings of the International Conference on Innovative Computing & Communication (ICICC), 2024): https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4832572

[7] Aliyu, A. A., & Liu, J. (2023). Blockchain-Based Smart Farm Security Framework for the Internet of Things. *Sensors*, *23*(18), 7992. https://doi.org/10.3390/s23187992